Open Critical Mapping Pre-Workshop

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What is R?

R is a programming language. However, unlike other programming languages, "it was not created for software engineers for software development. Instead, it was developed by statisticians as an interactive environment for data analysis."(Irizarry 2021) It provides an environment where uses can manipulate data, run computations, and create graphics. R performs as a calculator for basic mathematical operations. However, it can provide various tools for statistical analysis such as linear and nonlinear modeling, classical statistical tests, time-series analysis, classification, and clustering, etc. Moreover, you can conduct interdisciplinary research with text analysis, geographic information system (GIS) analysis, and image analysis, etc. R recently has developed its capacity for data mining and machine learning.(Watson 2021) R also offers excellent graphic tools for data visualization.

> "Companies as diverse as Google, Pfizer, Merck, Bank of America, the InterContinental Hotels Group and Shell use R. [...] The great beauty of R is that you can modify it to do all sorts of things, said Hal Varian, cheif economist at Google. And you have a lot of prepackaged stuff that that is already available, so you are standing on the sholders of giants." (Vance 2009)

Advantages of R

Why we use R?

- Widely-used in academia and industries
- Open-source and free
- Power and flexibility
- Graphical capabilities
- A way to learn basic programming (Rosenberg 2020)

Installing R

How to download R?

Downloads for Windows, Mac, and Linux systems are all available at this site: https://cloud.r-project.org/ You can select the correct installation file for your operating system. For Windows, select the "base" installation. For MacOSX, choose "R 4.6.1.pkg".

If you use a Google Chromebook, you can access to R studio cloud at https://rstudio.cloud/.

Installing R Studio

R Studio is a good interface to use R. It provides you within one window a text editor, graph viewer, the R console, and access to help files. You can install R Studio at https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads.

When you open R Studio, it automatically find your R installation. You do not need to open R additionally. In R Studio, click on `File - New - R script`. This opens a new blank document where you will write all your commands. The R script does not run the commands automatically. You should **run** the commands in the R Console.

The R console is located in the lower left corner of the R Studio window (as a basic setting - However, you can change this setting: `Tools - Global Options - Pane Layout`). This is where your commands will be executed. When using R Studio, you have two options: (1) write commands in the text editor (located in the upper left corner) or (2) directly enter into the R console. I highly recommend you to use the text editor. You can see the history of code you have saved. To run your code from the text editor, simply highlight the code and press **Run**. In Windows, ctrl+Enter works as a shortcut. (Rosenberg 2020)

In R Studio window, the top-right panel has the workspace tab which shows the list of R objects (data frame, variables, models, etc.) you created during your R session. The history tab shows the history of all previous commands.

The bottom-right panel has three tabs:

- Files tab shows files in your current working directory.
- Plots tab shows the history of plots/graphs you created. From this tab, you can export a plot to a PDF or an image files and save them in your computer.
- Package tab shows external R packages available on your system. If a certain package is checked, the package is loaded in R. (I will explain more in the next lab session.) ("Running RStudio," n.d.)

Important R Tips

1. All commands in R are case sensitive. R Studio provides some spell check but you always should double-check your commands.
2. The pound sign (#) is the symbol for comments. Anything you place on a single line behind this symbol will not be communicated to R, even if you place it in the command line. This can be useful for making notes to yourself or saving code that you do not want to run for some reason.
3. The up and down arrow keys in the R console, when used on the command line, will allow you to scroll back and forth through commands you have already executed in a single session.
4. **NEVER** overwrite your original data files.
5. **ALWAYS** save your R scripts.
6. When you quit a session in R Studio, R will ask you if you want to save your workspace image. The best answer is "No." You simply need to save your scripts. Once you do so, this is not something you will need to do.
7. If you get stuck in R, **Google** is the best source of help.

Working Directories in R

> "The working directory is just a file path on your computer that sets the default location of any files you read into R, or save out of R. If you ask R to import a dataset from a text file, or save a dataframe as a text file, it will assume that the file is inside of your working directory."

> "You can only have one working directory active at any given time. The active working directory is called your current working directory." (Phillips 2018)

Every single time whenever you use a new R session, you **SHOULD set up your working directory**.

To see your current working directory:

```
getwd()
```

```
## [1] "C:/Users/jjang/Dropbox/[Research]Voting/professional development workshop/mapping_with_census"
```

To change your working directory to the following path:

```
setwd("C:/Users/jjang/Dropbox/[Research]Voting/professional development workshop/mapping_with_census")
```

For **Windows** users, note that the slashes in this file path run the **opposite** direction of what you would copy and paste from your file explorer. Windows uses backslashes for file paths.(Watson 2021)

Since we are using this R Studio, there is one easy way to set up your working directory. Click `Session` - `Set Working Directory` - `Choose Directory` - select the file - `open`.

Installing Packages

We need to know the package name and follow this command: `install.packages("PACKAGENAME")` The console will display a variety of messages based on our operating system and then display "successfully unpacked and checked." Let's install a new package called "tidyverse" by this command: `install.packages("tidyverse")`

There is another way to install new packages. In R Studio, click the `Packages` tab at the bottom right panel. Afterwards, click another tab, `Install`. Once a new window pops up, type the package's name in the section called `packages`. If that package is enlisted in CRAN package repository, the name will automatically show up. Select the name and click `Install`.

Using Packages

Installing packages does not automatically make R use them. Every time you want to use the package, you need to load it. Let's load the new package, **tidyverse**:

```
library(tidyverse)
```

What happens if we try to use a package that we have not installed yet? R will tell us with the error message. R Studio also gives you a warning message (from your script) before you attempt to load the package.

Objects

Objects are the things we make in R. Each object has a variety of attributes such as a name, class, dimension, etc. Objects are assigned their value(s) with an arrow `<-` The values can be a single number, a set of numbers, the code for a graph, or an entire set of data.

With a few exceptions, R does not care what we call our objects.

1. Names should start with a letter (NOT a number).
2. Names should only contain letters, numbers, underscores(_), and periods.
3. Some things cannot be used as names: break, else, FALSE, for, function, if, Inf, NA, NaN, next, repeat, return, TRUE, while.
4. Some things should not be used as names such as any special characters, the names of functions, and T and F (they are often used as shorthand for true and false).
5. R is case sensitive. There is zero forgiveness for typos!!!

Example 1: Creating a Simple Object

```
x <- 5
x
```

```
## [1] 5
```

- `x` is the object name (the box label)
- `5` is the value stored inside (the content)

Example 2: Objects Can Store Many Types of Data

```
name <- "Morehouse College"     # a character object
numbers <- c(2, 4, 6, 8, 10)    # a numeric vector object
is_greater <- x > 3             # a logical object (TRUE/FALSE)
```

You can use the `class()` function to check what type of data an object stores:

```
class(name)
```

```
## [1] "character"
```

```
class(numbers)
```

```
## [1] "numeric"
```

```
class(is_greater)
```

```
## [1] "logical"
```

Example 3: Objects Can Be Combined to Create New Ones

```
y <- x + 5
y
```

```
## [1] 10
```

Example 4: Objects Stay in Your "Environment"

When you create an object, it appears in the Environment pane (usually at the top right of RStudio). This tells you what objects exist in your current R session and what they contain.

You can remove them if you do not need them anymore:

```
rm(x)              # Removing object x
rm(list = ls())    # Remove everything at once - Be careful!
```

U.S. Census API key

This workshop will support R users with the basics of mapping and spatial analysis with the U.S. Census data using the `tidycensus()` package. Additional information can be found on these slides here.

To access the census data for mapping, please sign up for an API key using this link. You should receive an API key in your email. Please keep the API key for next Monday/Wednesday workshop and do not share with anyone.

References

Irizarry, Rafael A. 2021. *Introduction to Data Science*. https://rafalab.github.io/dsbook/getting-started.html.

Phillips, Nathaniel D. 2018. *YaRrr! The Pirate's Guide to r*. https://bookdown.org/ndphillips/YaRrr/the-working-directory.html.

Rosenberg, Drew. 2020. *Political Data Analysis: Syllabus*.

"Running RStudio." n.d. Statistical Tools for High-Throughout Data Analysis (STHDA). http://www.sthda.com/english/wiki/running-rstudio-and-setting-up-your-working-directory-easy-r-programming.

Vance, Ashlee. 2009. "Data Analysts Captivated by r's Power." New York Times. 2009.

Watson, K. Anne. 2021. *Introduction to r*.